SparkStart & SparkBuddy: Round 1 Project Proposal

Small steps. Real progress. • Track: LLMs & RAG (The Task Sequencer)

1. Team Details

Field	Details (Please fill in)
Team Name	[Insert Team Name Here]
Team Leader	[Insert Team Leader Name Here] • Email: [Insert Email]
Team Member 2	[Insert Member 2 Name Here] • Email: [Insert Email]
Team Member 3	[Insert Member 3 Name Here] • Email: [Insert Email]
Team Member 4	[Insert Member 4 Name Here] • Email: [Insert Email]
Institution / University	[Insert University / College Name Here]
Primary Track	LLMs & RAG — The Task Sequencer

2. Brief About the Problem and Idea (Abstract)

SparkStart is an adaptive, low-pressure task sequencing system paired with SparkBuddy, a cross-application floating desktop companion. Designed for university students facing executive dysfunction and task initiation paralysis, SparkStart replaces intimidating 50-step checklists with evidence-based, 2-minute starter actions calibrated to self-reported energy levels. Utilizing a grounded Retrieval-Augmented Generation (RAG) architecture anchored in cognitive load and behavioral psychology literature, the system bridges the gap between intention and performance directly within active desktop workspaces such as VS Code. SparkStart rejects punitive streaks, aggressive alarms, and diagnostic overreach in favor of genuine user autonomy, local-first SQLite privacy, and guilt-free restorative pacing.

3. Proposed Solution and Unique Selling Propositions (USPs)

The Solution: SparkStart addresses the psychological hurdle of task initiation rather than enforcing rigid timetables. Instead of exposing students to long to-do backlogs that induce anxiety freeze, SparkStart isolates a single, observable, 2-to-5 minute starting action supported by a floating desktop companion (SparkBuddy) at the point of performance.

• **1. Single-Target Scaffolding:** Eliminates working memory saturation by displaying only one active micro-action.

• **2. Dynamic Energy Pacing:** Calibrates tasks to self-reported capacity (Low, Medium, High, Unsure) without guilt or shame.

• **3. Instant 'Too Big' Shrink Engine:** Decomposes resistant tasks recursively down to 60-second friction-free contact actions.

• **4. Point-of-Performance Companion:** Floats on top of VS Code and web browsers without stealing keyboard focus.

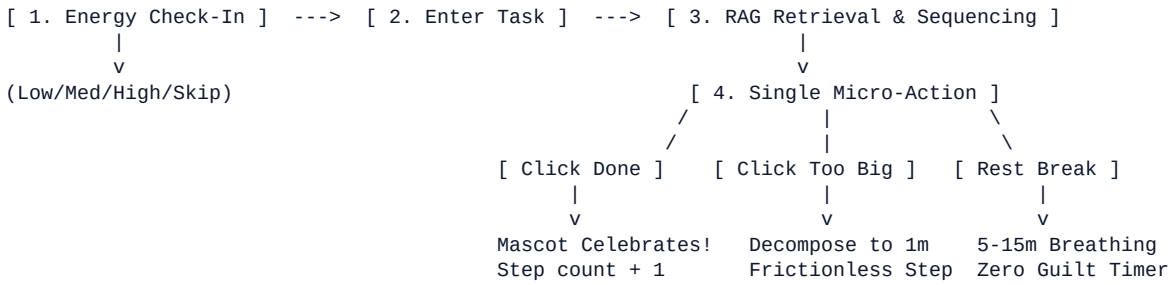
• **5. Evidence-Grounded RAG Pipeline:** Recommendations cite genuine psychological findings (Gollwitzer, Sweller, Barkley).

• **6. Local-First Privacy Sovereignty:** Operates 100% offline out-of-the-box with local SQLite persistence and explicit AI consent.

4. Key Features

Adaptive Task Sequencer	Generates single-action starter commitments with duration tags, manageability rationale, and gentle alternatives.
SparkBuddy Desktop Mascot	Electron companion with original animated sprout SVG displaying 5 emotive states (Idle, Encouraging, Thinking, Resting, Celebrating).
'Too Big' Action Shrink	One-click granular decomposition that immediately dissolves perfectionist freeze.
Guilt-Free Rest Timer	Configurable 5, 10, or 15-minute recovery intervals with soothing breathing guide and non-enforced return to work.
Grounded Evidence Drawer	Transparent literature citations showing peer-reviewed cognitive science foundations for each micro-step.
Privacy & Sovereignty	Instant JSON data export and permanent local SQLite database purge.

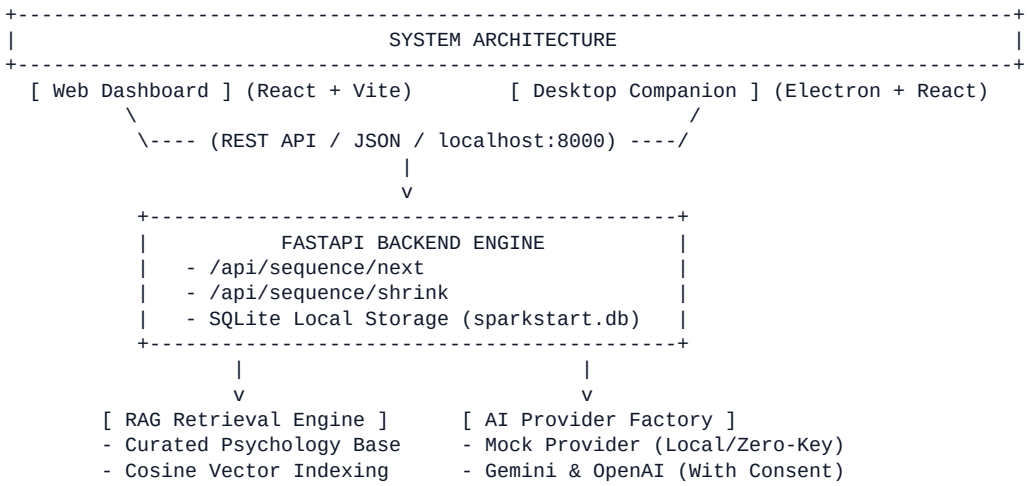
5. Process Flow / Use-case Diagram



6. Technical Track and Approach (LLMs & RAG)

Primary Track: LLMs & RAG — The Task Sequencer.
Methodology: Rather than using LLMs to hallucinate arbitrary multi-step schedules, SparkStart anchors generation in an indexed knowledge base of peer-reviewed behavioral research. Chunks are retrieved via token-boosted cosine similarity. The context is fed to the provider abstraction (Gemini / OpenAI / Mock), which validates structured JSON matching strict Pydantic schemas. If external networks are absent or remote AI is disabled, the local deterministic sequencer responds in under 5ms.

7. Architecture Diagram



8. Technologies to Be Used

Component	Technology Stack
Web Dashboard	React 18, TypeScript, Vite, Tailwind CSS, Lucide Icons
Desktop Companion	Electron 33, React 18, TypeScript, ContextBridge IPC
AI / Sequencer Backend	Python 3.14, FastAPI, Uvicorn, Pydantic v2
RAG & Search	Lightweight Cosine Vector Search, Curated JSON Knowledge Base
Local Database	SQLite3 (Local-first, relational schema)
Testing & Quality	Pytest, TestClient, tsc, Vite build bundler

9. Privacy, Ethics and User Experience

• **User Autonomy:** No punitive streaks, overdue shame alerts, or forced digital lockouts.

• **Zero Involuntary Transmission:** Operates 100% offline out-of-the-box. Remote AI requires opt-in consent.

• **Zero Intrusive Tracking:** No webcam access, gaze tracking, keystroke logging, or background surveillance.

• **Complete Data Sovereignty:** Immediate JSON data export and permanent local SQLite purge options.

• **Non-Clinical Positioning:** Designed as supportive scaffolding; does not claim to diagnose or treat ADHD or anxiety.